

• Now we shall verify whether $x = A^{-1}b$ is really a solution.

$$\begin{aligned} Ax &= A(A^{-1}b) = (AA^{-1})b \quad (\text{Theorem 2(a)}) \\ &= I_n b \quad (\text{By defn. of inverse}) \\ &= b \quad (\text{Theorem 2(e)}). \end{aligned}$$

This is the existence part. In the first part, we proved uniqueness.

$\therefore x = A^{-1}b$ is the unique solution $\square$

The formula in Theorem 5 is seldom used to solve an equation $Ax=b$ numerically because row reduction of $[A \ b]$ is nearly always faster. (Row reduction is usually more accurate, too, when computations involve rounding off numbers.) One possible exception is the 2×2 case.

Example 4:

system

$$\begin{aligned} 5x_1 + 4x_2 &= 5 \\ 5x_1 + 6x_2 &= 7 \end{aligned}$$

Solution:

This system is equivalent to $Ax=b$, so

$$x = A^{-1}b = \begin{bmatrix} -3 & 2 \\ 5/2 & -5/2 \end{bmatrix} \begin{bmatrix} 5 \\ 7 \end{bmatrix} = \begin{bmatrix} 5 \\ -3 \end{bmatrix} \square$$

Theorem 6:

a. If A is an invertible matrix, then A^{-1} is invertible and

$$(A^{-1})^{-1} = A.$$

b. If A and B are $n \times n$ invertible matrices, then so is AB , and the inverse of AB is the product of the inverses of A and B in the reverse order. That is,

$$(AB)^{-1} = B^{-1}A^{-1}.$$

c. If A is an invertible matrix, then so is A^T , and the inverse of A^T is the transpose of A^{-1} . That is, $(A^T)^{-1} = (A^{-1})^T$.

Proof:

a) By the defn. of invertible matrix, there exists a unique matrix A^{-1} such that $A^{-1}A = AA^{-1} = I_n$

• Let there exists an $n \times n$ matrix B such that $A^{-1}B = I_n$.

$$\begin{aligned} A \cdot (A^{-1}B) &= A \cdot I_n \Rightarrow (A \cdot A^{-1})B = A \cdot I_n \quad (\text{Theorem 2(a)}) \\ &\Rightarrow I_n \cdot B = A \cdot I_n \quad (\text{By defn. of inverse}) \end{aligned}$$